Department of Electrical and Computer Engineering
North South University

Senior Design Project

Arsenic Skin Detection Using Machine Learning

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Spring, 2026

LETTER OF TRANSMITTAL

April, 2026
To
Dr. Mohammad
Abdul Matin
Chairman,

Department of Electrical and Computer Engineering

North South University, Dhaka

Subject: Submission of Capstone Project Report on "**Arsenic Skin Detection Using Machine Learning"**

Dear Sir,
With due respect, we are pleased to submit our Capstone Project Report titled "Arsenic Skin Detection using Machine Learning" as a partial requirement of our BSc in Computer Science and Engineering program. This project allowed us to explore how deep learning can assist in identifying arsenic-affected skin lesions from images a problem that holds immense importance for public health in Bangladesh.

Throughout the project, we have tried our best to apply our theoretical knowledge to real-world data and develop a meaningful solution. We sincerely hope that this report meets your expectations and provides a clear understanding of our work, methodology, and findings.

We would be grateful if you kindly accept this report and evaluate our work.

Sincerely yours,

Tanushree Das
ID: 2212225042
CSE Department

Ashraful Islam
ID: 2212669042
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APPROVAL

This is to certify that **Tanushree Das (2212225042)**, and **Ashraful Islam (2212669042)** from the Department of Electrical & Computer Engineering at North South University have completed the Capstone Project titled:

"Arsenic Skin Detection using Machine Learning"

The project has been completed under my supervision and is accepted as satisfactory for partial fulfillment of the BSc degree requirements.

Supervisor's Signature

…….

Dr. Mohammad Shifat-E-Rabbi [MSRb]

Assistant Professor

Department of Electrical and Computer Engineering

North South University

Dhaka, Bangladesh.

Chairman's Signature

…….

Dr. Mohammad

Abdul Matin [mtn]

Professor

Department of Electrical and Computer Engineering

North South University

Dhaka, Bangladesh.

DECLARATION

We hereby declare that the work presented in this report is entirely our own. No part of this report has been submitted elsewhere for any academic degree or diploma. All materials used from other published or unpublished sources have been properly referenced.

We also acknowledge that all project-related information is confidential and will not be disclosed without the permission of our supervisor.

Students' Names & Signatures:

1. Tanushree Das

2. Ashraful Islam

ACKNOWLEDGEMENTS

First and foremost, we express our deepest gratitude to our supervisor, Dr. Mohammad Shifat-E-Rabbi for his continuous guidance, valuable feedback, and encouragement throughout the entire project. His support made this complex project possible.

We would also like to thank the Department of Electrical and Computer Engineering, North South University, for providing the facilities and resources necessary to carry out our research.

We are grateful to our friends and classmates who encouraged us and offered meaningful suggestions during the development process. Finally, we thank our families for their unconditional support, patience, and motivation throughout this journey.

ABSTRACT

Arsenic Skin Detection using Machine Learning

Arsenic contamination in groundwater is a critical public health crisis in Bangladesh, causing severe skin lesions that can lead to long-term complications including skin cancer. This project presents an end-to-end intelligent healthcare system for early detection of arsenic-affected skin lesions by combining machine learning with an accessible web-based platform. Machine learning models were developed using the ArsenicSkinImageBD dataset containing 8,892 balanced images, preprocessed through a standardized pipeline involving resizing and normalization. A baseline Convolutional Neural Network (CNN) achieved a classification accuracy of 85.44%, while a ResNet50-based transfer learning model was also implemented; although initial attempts faced architectural issues, the flaws were systematically analyzed and a redesigned model was proposed for future improvement. To make the diagnostic capability accessible to real users, a full-stack web application was developed incorporating user registration and login, a structured patient information form for clinical data collection, an AI-powered chatbot for user guidance, and a review and feedback system for continuous improvement. This integrated system demonstrates the feasibility of combining deep learning with modern web technologies to build a practical diagnostic tool, establishing a strong foundation toward a high-accuracy, real-world healthcare application addressing a pressing public health concern in Bangladesh.

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Chapter 1 Introduction

1.1 Background and Motivation

Arsenic-contaminated groundwater is a major health problem in Bangladesh, especially in rural areas where millions still drink unsafe water. Long-term exposure causes visible skin issues like dark spots, rough patches, and lesions. Detecting these early is critical, but many clinics lack trained dermatologists, causing symptoms to be frequently missed or diagnosed too late.

With smartphones now widespread, an automatic skin-image detection tool could help health workers identify arsenic-related conditions more easily. Modern machine learning, especially CNN-based image analysis, offers a promising and cost-effective approach to achieve this. However, an accurate model alone is insufficient for real-world impact. This motivated us to develop a supporting web application featuring user registration and login, a structured patient information form, an AI-powered chatbot for user guidance, and a review and feedback system.

Our project aims to build a simple, accessible, and reliable system that can support early detection of arsenic skin symptoms and improve public health in underserved communities.

1.2 Purpose and Goal of the Project

This project aims to build a machine learning model that classifies skin images as "infected" or "not infected" to support early detection of arsenic-related skin conditions. A full pipeline was developed, including a baseline CNN and a ResNet50-based transfer learning model, with results compared against the ArsenicNet benchmark of 97.69%. Alongside the ML model, a web application was developed featuring user registration and login, a patient information form, an AI-powered chatbot, and a review and feedback system. Together, these components create a complete, accessible diagnostic platform to support low-cost early screening for arsenic-related skin issues in rural communities across Bangladesh.

1.3 Organization of the Report

The rest of this report is structured as follows:

· **Chapter 2** reviews previous research on arsenic skin detection and deep learning techniques.

· **Chapter 3** explains the dataset, preprocessing pipeline, and machine learning models used.

· **Chapter 4** presents the experiments, model results, and analysis.

· **Chapter 5** describes the design and development of the web application, including user registration, login, patient information form, AI-powered chatbot, and review and feedback features.

· **Chapter 6** covers the system architecture, technology stack, and implementation details of the web platform.

· **Chapter 7** discusses the societal, health, and environmental impacts of this project.

· **Chapter 8** includes project planning and budget estimation.

· **Chapter 9** outlines the complex engineering problems and activities addressed.

· **Chapter 10** concludes the report with limitations and future improvements.

Chapter 2 Research Literature Review

2.1 Existing Research and Limitations

Researchers are paying more attention to early detection of arsenic-related skin problems because of their serious impact on public health. Recent studies have used machine learning and deep learning to classify skin abnormalities. One key resource is the ArsenicSkinImageBD dataset, which contains over 8,000 labeled images from Bangladesh. Another major contribution is the ArsenicNet model a hybrid Xception-Inception network that reached 97.69% accuracy and set a strong benchmark for this field. Other dermatology studies also show that CNNs are highly effective for spotting subtle skin-related patterns.

Identified Limitations in Existing Research
A closer review of the literature highlights several gaps:

1. **Complex architectures dominate the field.**
   Many high-performing models (like ArsenicNet) use hybrid or extremely deep networks, making them harder to deploy on low-power devices used in rural healthcare centers.
2. **Limited comparison between simple and advanced models.**
   Many studies jump directly to advanced architectures rather than systematically comparing baseline models with transfer learning approaches.
3. **Lack of model explainability.**
   Most research does not include feature visualization or explainable AI components, which are critical for medical trust and transparency.
4. **Few studies address debugging and model failure cases.**
   Almost no research discusses model instability, incorrect implementations, or effects of improper preprocessing—issues we encountered during our initial ResNet50 attempt.

These findings strongly motivated our project to evaluate both a simple CNN baseline and a ResNet50 transfer learning model, helping determine whether a lightweight model can still achieve competitive accuracy.

Chapter 3 Methodology

3.1 System Design

.**The project follows a two-part pipeline covering both machine learning and web application development.**

For the machine learning component, we used the ArsenicSkinImageBD dataset containing 8,892 images, which were resized, normalized, and labeled as infected (1) or not infected (0). The data was split 80/20 for training and testing. A baseline CNN and a ResNet50-based transfer learning model were built, evaluated using accuracy and other metrics, and compared against the 97.69% ArsenicNet benchmark.

For the web application component, a full-stack platform was developed to make the diagnostic tool accessible to real users. The system includes a user registration and login page for secure access, a structured patient information form for collecting clinical details, an AI-powered chatbot to guide users and answer basic queries, and a review and feedback option for continuous improvement. Together, both components form a complete end-to-end system that integrates intelligent skin lesion detection with an accessible and user-friendly web interface, aimed at supporting early diagnosis of arsenic-related skin conditions in underserved communities.

Software Components

Since this is a machine learning–based software project integrated with a web application, the implementation requires:

- ➔ For the machine learning component: Python, TensorFlow/Keras, NumPy,Open CV, Matplotlib, and Google Colab GPU.
- ➔ For the web application component: HTML, CSS, and JavaScript for frontend development, Ollama for powering the AI-based chatbot, and Jupyter Notebook for model experimentation and integration support.

TABLE I. A SAMPLE SOFTWARE TOOLS TABLE

Tool	**Function**	**Alternative Options**	**Why Selected**
Python	**Core programming language**	**C++, MATLAB**	**Most popular ML language, easy to debug**
TensorFlow / Keras	**Build and train CNN & ResNet models**	**PyTorch**	**Easy Sequential API, Colab support**
NumPy	**Numerical operations, arrays**	**Pandas**	**Faster manipulation for images**
PIL / OpenCV	**Image processing**	**skimage**	**Widely used and stable**
Matplotlib	**Plotting accuracy/loss curves**	**Seaborn**	**Simple and customizable**
Google Colab	**GPU training environment**	**Kaggle Notebook**	**Free GPU and easy collaboration**
Jupyter Notebook	**Model experimentation and integration support**	**VS Code**	**Interactive coding and easy visualization**
HTML	**Frontend structure and layout**	**Vue**	**Simple, lightweight, no framework needed**
CSS	**Styling and responsive design**	**Tailwind**	**Full control over custom UI design**
JavaScript	**Frontend interactivity and logic**	**TypeScript**	**Widely supported, no compilation needed**
Ollama	**AI-powered chatbot integration**	**OpenAI API**	**Local LLM support, free and private**

3.2 Software Implementation

- **Data Prep (Tanushree Das):** Loaded 8,892 images, resized to 128×128, normalized, and split into train/test sets.
- **Baseline CNN (Ashraful):** Simple CNN with Conv → Pool → Dense → Dropout → Output, compiled with Adam.
- **ResNet50 Model (Ashraful):** Pre-trained ResNet50 with frozen base, GAP → Dense → Dropout → Dense(2, softmax).
- **User Authentication(Tanushree Das)**: Register and Login pages built with HTML → CSS → JavaScript → Form Validation → Session Handling.
- **Patient Information Form(Ashraful)**: Input Fields → Dropdown → Text Area → Submit Button → Data Capture, structured for clinical data collection.
- **AI-Powered Chatbot(Tanushree Das)**: User Input → Ollama LLM → Response Generation → Display Output, integrated for real-time user guidance.
- **Review and Feedback System(Tanushree Das)**: Star Rating → Comment Box → Submit → Feedback Storage, designed for continuous platform improvement.
- **Frontend Development(Ashraful)**: HTML Structure → CSS Styling → JavaScript Logic → Responsive Layout → Browser Render.
- **Challenge:** Initial ResNet50 used UpSampling2D, causing distortion; fixed by redesigning input layer.

Chapter 4 Investigation/Experiment, Result, Analysis and Discussion

- **Setup:** Experiments ran on Google Colab GPU. Dataset had 8,892 images (infected vs. not infected, balanced). Images resized to 128×128, normalized, and split 80/20.
  Labels: 1 = infected, 0 = not infected. Alongside the machine learning experiments, a full-stack web application was developed and tested locally using HTML, CSS, JavaScript, and Ollama for chatbot integration.
- **Baseline CNN:**
- Layers: Conv → Pool → Conv → Pool → Flatten → Dense → Dropout → Dense(2, softmax)
- Training: 10 epochs, Adam, sparse_categorical_crossentropy
- Results**:** 85.44% test accuracy. Misclassified some subtle infected cases. Mild overfitting observed.
- **ResNet50 (Initial):**
- Transfer learning with frozen ResNet50 + GAP + Dense layers
- Problem**:** UpSampling2D distorted images; input size mismatch; trainable layers misconfigured → accuracy dropped to 77%.
- **ResNet50 Fix Plan:**
- Remove UpSampling2D, use correct input (128×128×3), freeze base, add classification head, retrain with augmentations
- Expected: Higher accuracy, better generalization
- **Web Application — Implementation and Testing**
- **User Authentication:** Registration and login tested for form validation, error handling, and session management. Edge cases such as empty fields and mismatched passwords handled correctly.
- **Patient Information Form:** Clinical data fields (name, age, location, symptoms) captured and displayed accurately upon submission.
- **AI-Powered Chatbot:** Ollama-based chatbot tested for response quality and latency. Successfully handled basic health queries and guided users in real time.
- **Review and Feedback System:** Star rating and comment submission tested across multiple interactions. Feedback stored and displayed correctly.
- **Frontend Responsiveness:** All pages verified for responsive layout across different screen sizes using HTML and CSS.

Discussion:

- CNN works, but ResNet50 can do better if properly set up.
- Data quality and correct architecture are key.
- Benchmark (ArsenicNet, 97.69%) shows room for improvement.
- Web application features all functioned as intended, confirming real-world feasibility.
- **Future:** mobile app development, and integration of Gemini API for enhanced AI-powered chatbot capabilities.

Chapter 5: Web Application Design and Development

5.1 **User Registration and Login**

- Secure authentication system with registration and login pages
- Form validation and error handling implemented
- Session management ensures safe and reliable platform access

5.2 **Patient Information Form**

- Structured form collects name, age, location, and symptom description
- Data captured and stored accurately upon submission
- Supports diagnostic assessment and clinical data collection

5.3 **AI-Powered Chatbot**

- Integrated using Ollama as local LLM backend
- Guides users through the platform and answers basic health queries
- Operates in real time with no internet-based API required

5.4 **Review and Feedback System**

- Users can submit star ratings and written comments
- Feedback stored and displayed correctly
- Enables continuous improvement based on real user experience

Chapter 6: System Architecture and Implementation

6.1 **System Architecture and Technology Stack**

- Client-side architecture built with HTML, CSS, and JavaScript
- Ollama serves as local LLM backend for chatbot functionality
- No external server dependency in the current phase

6.2 **Frontend Implementation**

- All pages designed with HTML and CSS for clean, responsive layout
- JavaScript handles form validation, page logic, and dynamic interactions
- Tested across multiple screen sizes for responsiveness

6.3 **Chatbot Integration and Deployment**

- Ollama integrated locally to ensure data privacy
- Chatbot runs on user's machine with no external API calls
- Future deployment: live server hosting with Gemini API integration

Chapter 7 Impacts of the Project

7.1 Impact of this project on societal, health, safety, legal and cultural issues

- **Health:** Early detection of skin lesions helps prevent serious damage and encourages timely treatment.
- **Society:** AI tools assist rural healthcare workers and raise awareness about arsenic risks.
- **Safety:** Accurate detection protects vulnerable people and reduces risk of serious illness.
- **Ethics & Privacy:** Patient data must be protected; AI must be fair and transparent.
- **Culture:** Smart web screening makes testing easier and reduces stigma in rural communities.

7.2 Impact of this project on environment and sustainability

Though this project is software-based, it indirectly contributes to environmental sustainability:

Environmental Impact

- Arsenic is primarily an environmental contaminant.
- Early detection helps bring attention to contaminated sources linked to groundwater pollution.

Sustainability Considerations

- AI screening tools reduce dependency on physical health infrastructure.
- Long-term: Can assist NGOs and public health organizations in tracking arsenic patterns.
- Encourages sustainable healthcare solutions that operate with low resources.

This project demonstrates how software innovation can support environmental awareness and promote sustainable public health strategies.

Chapter 8 Project Planning and Budget

8.1 Project Planning

A structured timeline was followed to ensure smooth progress throughout the semester. The work was divided into phases: dataset acquisition, preprocessing, model design, baseline model training, advanced model implementation, debugging, evaluation, web application development, and report writing.

Below is the Gantt chart (standard BAETE representation):

Task	Week 1–2	Week 3–4	Week 5–6	Week 7–8	Week 9–10	Week 11–12	Week 13–14
Literature Review							
Dataset Collection & Preprocessing							
Baseline CNN Model & ResNet50 Model							
Debugging & Re-Training							
Web App Design (UI/UX)& User Auth & Patient Form							
Chatbot Integration & Review & Feedback System							
Testing, Debugging & Result Analysis							
Report Writing							

Figure 1. Gantt chart.

This is mainly a software project with no hardware needed. Most tools, like Google Colab, are free, but an estimated budget is listed below.

Item	**Quantity**	**Estimated Cost (BDT)**	**Notes**
Laptop / PC	1	0 (already owned)	Used for coding & experiments
Google Colab GPU	1	0	Free tier used
Internet Usage	~60GB	1,000	Dataset download, training
Cloud Storage (Google Drive)	15GB	0	Free
Software Tools (Python, TensorFlow)	-	0	Open-source
Languages(HTML / CSS / JavaScript)	-	0	Open-source, no license needed
Ollama (AI Chatbot)	1	0	Free, runs locally
Jupyter Notebook	-	0	Open-source
Report Printing & Binding	1	300	Final submission
Total Estimated Cost		**1,300 BDT**	

Figure 2. A sample budget table.

Chapter 9 Complex Engineering Problems and Activities

9.1 Complex Engineering Problems (CEP)

This chapter explains how the project satisfies BAETE-defined Complex Engineering Problems (CEP) and Complex Engineering Activities (CEA).

Attribute	How Our Project Addresses CEP
P1: Depth of Knowledge Required	Requires knowledge of deep learning, CNNs, transfer learning, and image preprocessing (K4–K6). Web development required HTML, CSS, JavaScript, and Ollama-based LLM integration.
P2: Range of Conflicting Requirements	Balancing model accuracy with computational efficiency was challenging. On the web side, balancing feature richness (chatbot, forms, feedback) with simplicity and accessibility posed design conflicts.
P3: Depth of Analysis Required	Detailed analysis of classification reports, validation curves, and ResNet50 debugging was needed. Web features were tested for response quality, form accuracy, and user feedback patterns.
P4: Familiarity with Issues	Handling image noise, lighting variation, and arsenic lesion features were key ML challenges. Web challenges included Ollama integration, session security, and form input edge cases.
P5: Applicable Standards/Codes	Data integrity, privacy, and fairness were followed as ethical guidelines. The web platform applied standard practices for secure login, patient data privacy, and responsible chatbot usage.
P6: Stakeholder Involvement	Stakeholders include patients, health workers, dermatologists, and researchers. The web platform directly serves patients and health workers through its interface, chatbot, and feedback system.
P7: Interdependence of Systems	Dataset handling, preprocessing, CNN, and ResNet50 pipelines are all interdependent. The web platform adds further interdependence through its frontend, chatbot, patient form, and ML model integration.

Table I demonstrates a sample complex engineering problem attribute.

9.2 Complex Engineering Activities (CEA)

TABLE II. A SAMPLE COMPLEX ENGINEERING PROBLEM ACTIVITIES TABLE

Attributes		**Addressing the complex engineering activities (A) in the project**
A1	Range of resources	Involves human resources, open-source tools (Python, TensorFlow, Ollama), Google Colab GPU, and web technologies (HTML, CSS, JavaScript) — all utilized at minimal cost.
A2	Level of interactions	Involves interactions between team members, dataset providers, and end users (patients, health workers). The web platform enables direct user interaction through login, patient forms, chatbot, and feedback features.
A3	Innovation	Employs ML-based skin lesion detection combined with an AI-powered chatbot (Ollama) and a web diagnostic platform — an innovative approach to address a real public health problem in Bangladesh.
A4	Consequences to society / Environment	Supports early detection of arsenic-related skin disease in rural communities, reducing health risks. The web platform improves healthcare accessibility with no environmental hardware footprint.
A5	Familiarity	Requires familiarity with CNN/ResNet50 architectures, ML pipelines, web development, and LLM integration. UN SDG #03: Good health; UN SDG #04: Quality education; UN SDG #10: Reduce inequality.

Chapter 10 Conclusions

10.1 Summary

The aim of this project was to develop a machine learning model capable of identifying arsenic-related skin lesions from images. Using the **ArsenicSkinImageBD** dataset of 8,892 images, the team built:

- A **Baseline CNN**
- An **Advanced ResNet50 Transfer Learning Model**
- A Full-Stack Web Application with login, registration, patient form, **AI-powered chatbot**, and review system

The baseline CNN achieved **85.44% accuracy**. The initial ResNet50 faced technical issues due to faulty upsampling and improper layer configuration, and a corrected strategy was established for the next iteration. The web platform was successfully developed and tested, confirming real-world feasibility.

10.2 Limitations

- The ResNet50 model was not fully optimized due to incorrect initial implementation.
- Training was performed on free GPU resources, limiting epochs and fine-tuning.
- No data augmentation was applied in the final experiment, affecting accuracy.
- The project did not include explainability tools like Grad-CAM, important for medical AI.
- Real-world testing with new images was outside the project scope.
- The web application is currently locally hosted and not yet deployed on a live server.
- The chatbot (Ollama) is limited to general guidance and not yet medically specialized.

10.3 Future Improvements

Future work can significantly enhance the system's performance:

1. **Fine-tune ResNet50**
    - Unfreeze last 20–30 layers for deeper learning
2. **Add explainability features (Grad-CAM)**
    - Increase trust for medical usage
3. **Train with more epochs or larger GPUs**
    - Achieve higher accuracy closer to ArsenicNet's 97.69% benchmark

4. **Deploy the web application on a live server for real-world public access**
5. **Integrate Gemini API to enhance chatbot intelligence and medical response quality**
6. **Build a mobile app prototype**
    - Allow rural health workers to upload photos for instant predictions

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